Genetic engineering & Medicine

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Objectives:

At the end of this chapter, it is expected;

- To describe the basics of Genetic engineering, and the method of rDNA creation
- To appreciate the applications of Genetic engineering in Medicine and therapeutics
- To describe the methods of production of transgenics (GMOs)
- To appreciate the applications of transgenics in medicine and therapeutics
- To describe the method of constructing Genomic and cDNA library and its uses.

What is genetic engineering?

- Generally refers to rDNA technology (procedures used to carry out genetic engineering)
  - It means altering the genes (deliberate modification) in a living organism to produce a Genetically Modified Organism (GMO) with a new genotype.

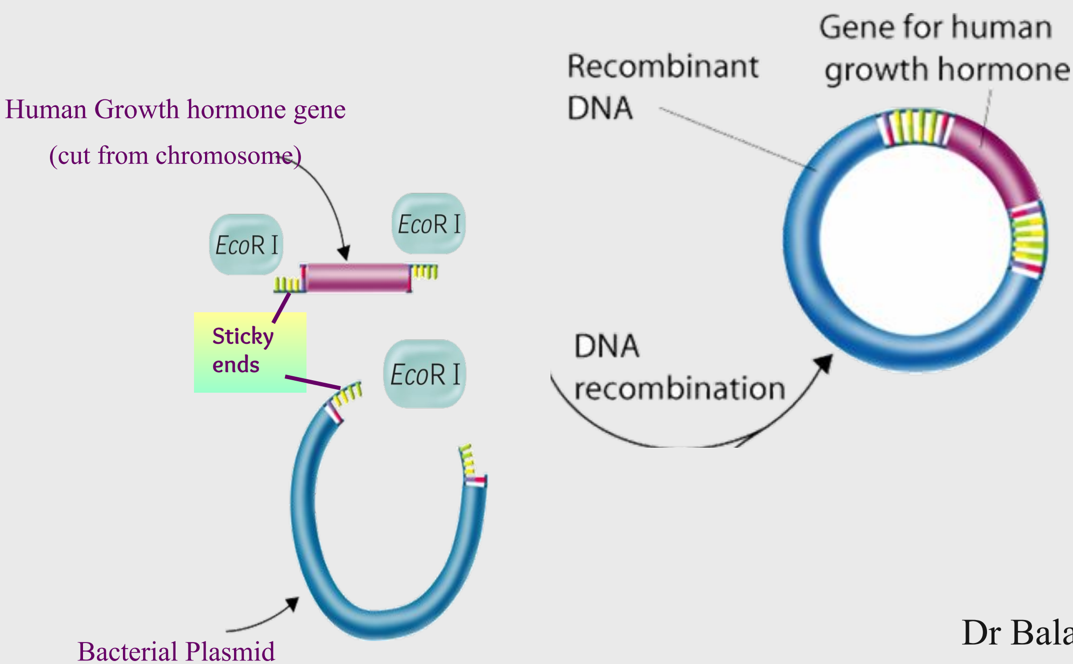
- Various kinds of genetic modification are possible:

1. Inserting a foreign gene from one species into another □ Forming a transgenic organism;
2. Altering an existing gene so that its product is changed; or
3. Changing gene expression so that it is translated more often or not at all.

Recombinant DNA (rDNA)

- A unique DNA molecule produced by joining together two or more DNA fragments, which are not normally associated with each other.

- DNA fragments are usually derived from different biological sources.



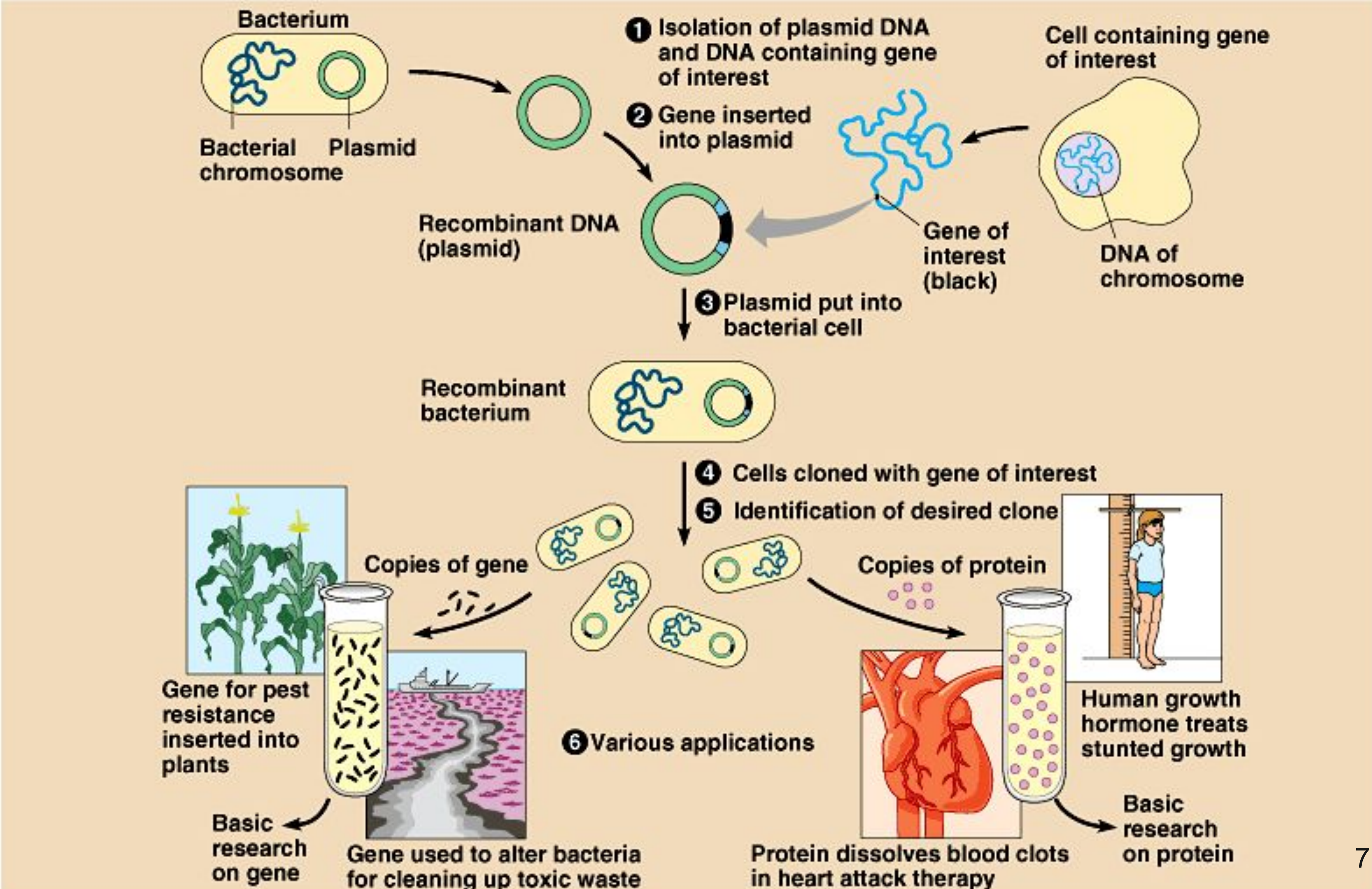
Applications of rDNA Technology

- Gene isolation/purification/synthesis
- Expression analyses (transcriptional and translational levels)
- Biochemistry/ Molecular modeling
- High throughput screening
- Mutagenesis (reverse genetics)
- Gene therapy
- Recombinant Vaccines
- Genetically modified crops
- Biosensors
- Monoclonal antibodies
- Xenotransplantation
- Bioremediation
- Production of next generation antibiotics
- Forensics
- Bioterrorism detection

Basic steps in genetic engineering

1. Isolate the gene of interest and a vector of choice
2. Integrate the gene of interest in to the vector to produce rDNA
3. Insert it in a host
4. Produce as many copies of the host as possible
5. Extraction of desired gene product
6. Separate and purify the product of the gene

Process of genetic engineering



GENETIC ENGINEERING APPLICATIONS IN MEDICINE

Genetic engineering and Medicine

- Plays a very important role in;
  - **Scientific research,**
  - **Diagnosis and treatment of disease**.
- Permits scientists and clinicians to:
  - **Identify new genes and the proteins they encode**
  - **To correct endogenous genetic defects**
  - **To manufacture large quantities of specific gene products such as hormones, vaccines, and other biological agents of medical interest**
    - Advantages include:
      - High purity
      - High specific activity
      - Steady supply
      - Batch-to-batch consistency

Genetic engineering and Medicine

- **Identification of mutations:**
  - People may be tested for the presence of mutated proteins that may be involved in the progression of breast cancer, retino-blastoma, and neurofibromatosis.
  - **Diagnosis of affected and carrier states for hereditary diseases:**
    - Tests exist to determine if people are carriers of the cystic fibrosis gene, the Huntington's disease gene, the Tay-Sachs disease gene, or the Duchenne muscular dystrophy gene.
- **Mapping of human genes on chromosomes:**
  - Scientists are able to link mutations and disease states to specific sites on chromosomes.
- **Transferring genes from one organism to another:**
  - People suffering from cystic fibrosis, rheumatoid arthritis, vascular disease, and certain cancers may now benefit from the progress made in gene therapy.
- **Isolation and alteration of genes:**
  - Once gene modification becomes successful, alteration of genes to produce a more functional protein than the endogenous protein may become possible, opening up the route of gene therapy.

Genetic engineering and Medicine

- **Performing structure and function analyses on proteins:**
  - Researchers may now employ rational drug design to synthesize drug compounds that will be efficacious and selective in treating disease.
- **Isolation of large quantities of pure protein:**
  - Insulin, growth hormone, follicle-stimulating hormone, as well as other proteins, are now available as recombinant products.
  - Physicians will no longer have to rely on biological products of low purity and specific activity from inconsistent batch preparations to treat their patients.

TRANSGENICS AND MEDICINE

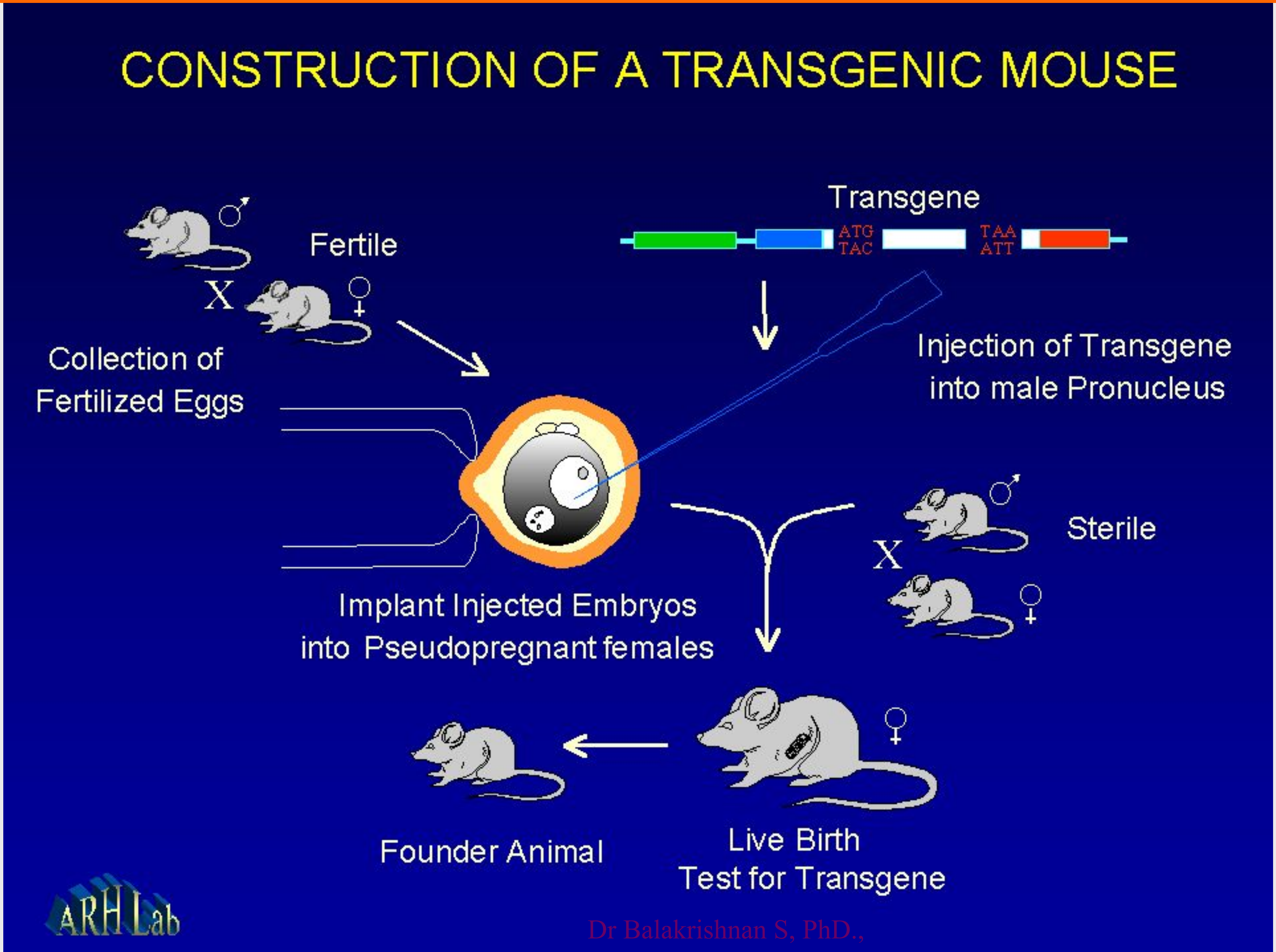
Transgenic Animals

- Animals that have their DNA manipulated.
- Animals that carry genes from other species.
  - For example, an embryo can have an extra, functioning gene from another source, artificially introduced into it, or
  - A gene introduced which can knock out the functioning of another particular gene in the embryo.
  - Animal biotechnology is the field to engineer transgenic animals.
- Useful as disease models and producers of substances for human welfare.
- Have already produced transgenic animals in mice, rats, rabbits, pigs, sheep, and cows.

Why are these animals being produced?

- Some transgenic animals are produced for specific economic traits.
  - e.g., Transgenic cattle were created to produce milk containing particular human proteins, which may help in the treatment of human emphysema.
- Other transgenic animals are produced as disease models
  - Animals genetically manipulated to exhibit disease symptoms so that effective treatment can be studied.
  - E.g., The OncoMouse® or the Harvard mouse, carrying a gene that promotes the development of various human cancers.

Transgenic Mice



How are transgenic animals produced?

- DNA microinjection
- Retrovirus-mediated gene transfer
- Embryonic stem cell-mediated gene transfer
- Sperm-mediated transfer
- Gene gun (Shutgun)
- Electroporation
- Lipofection

DNA microinjection

DNA microinjection is the main method used to create transgenic animals

- Introducing the transgene DNA directly into the zygote at an early stage of development.
- No vector required.

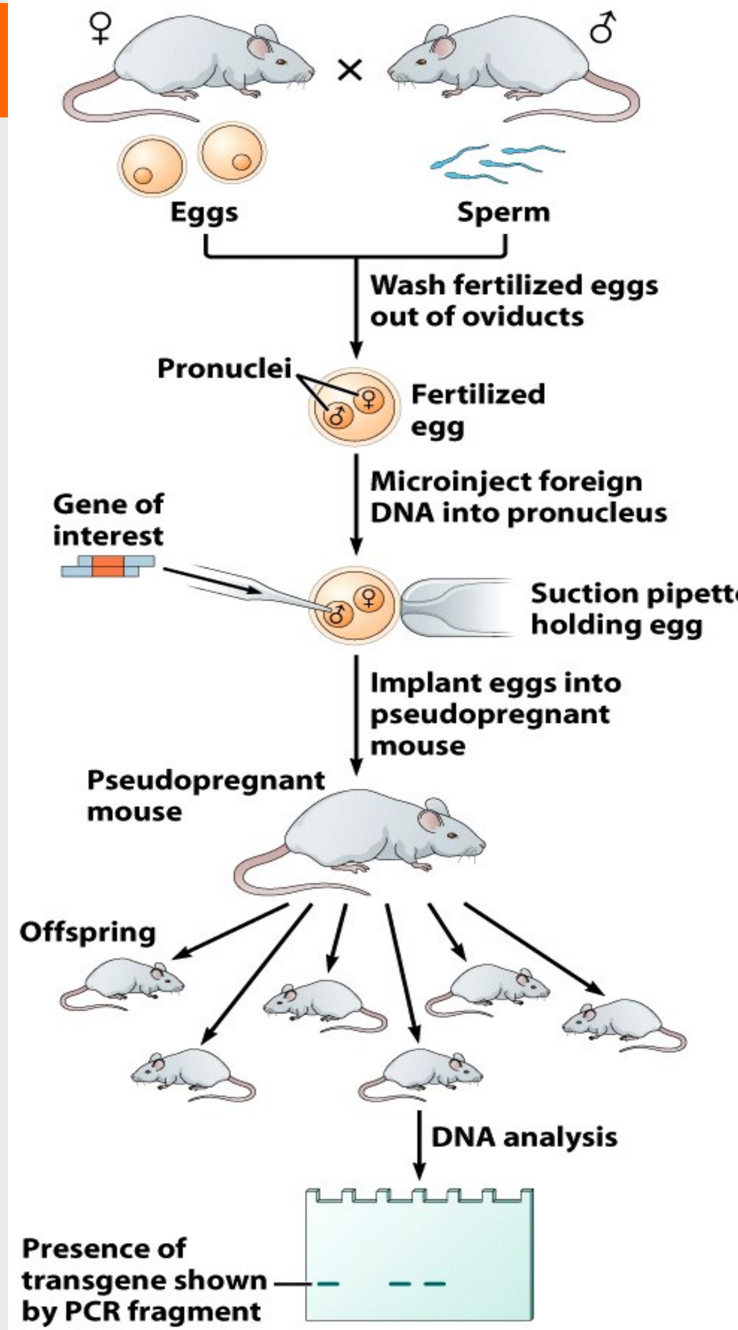
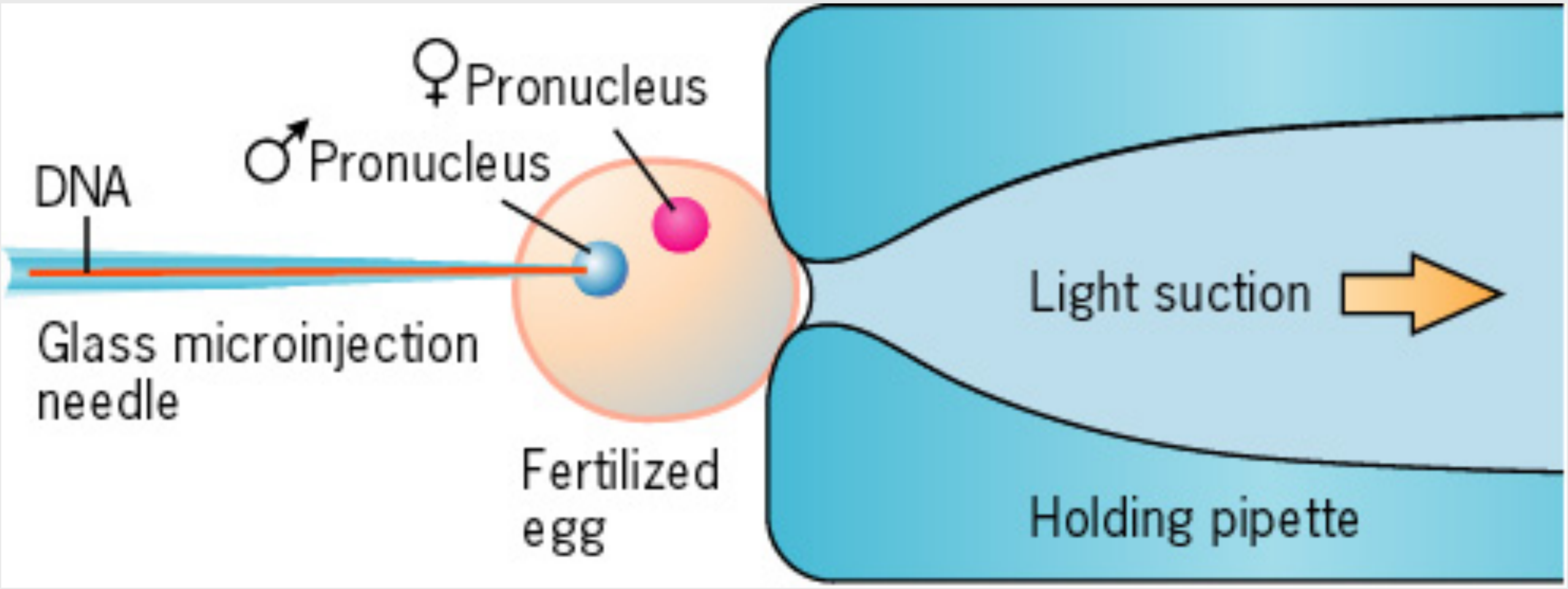


Figure 6-14
Genes and Genomics: A Short Course (3e)

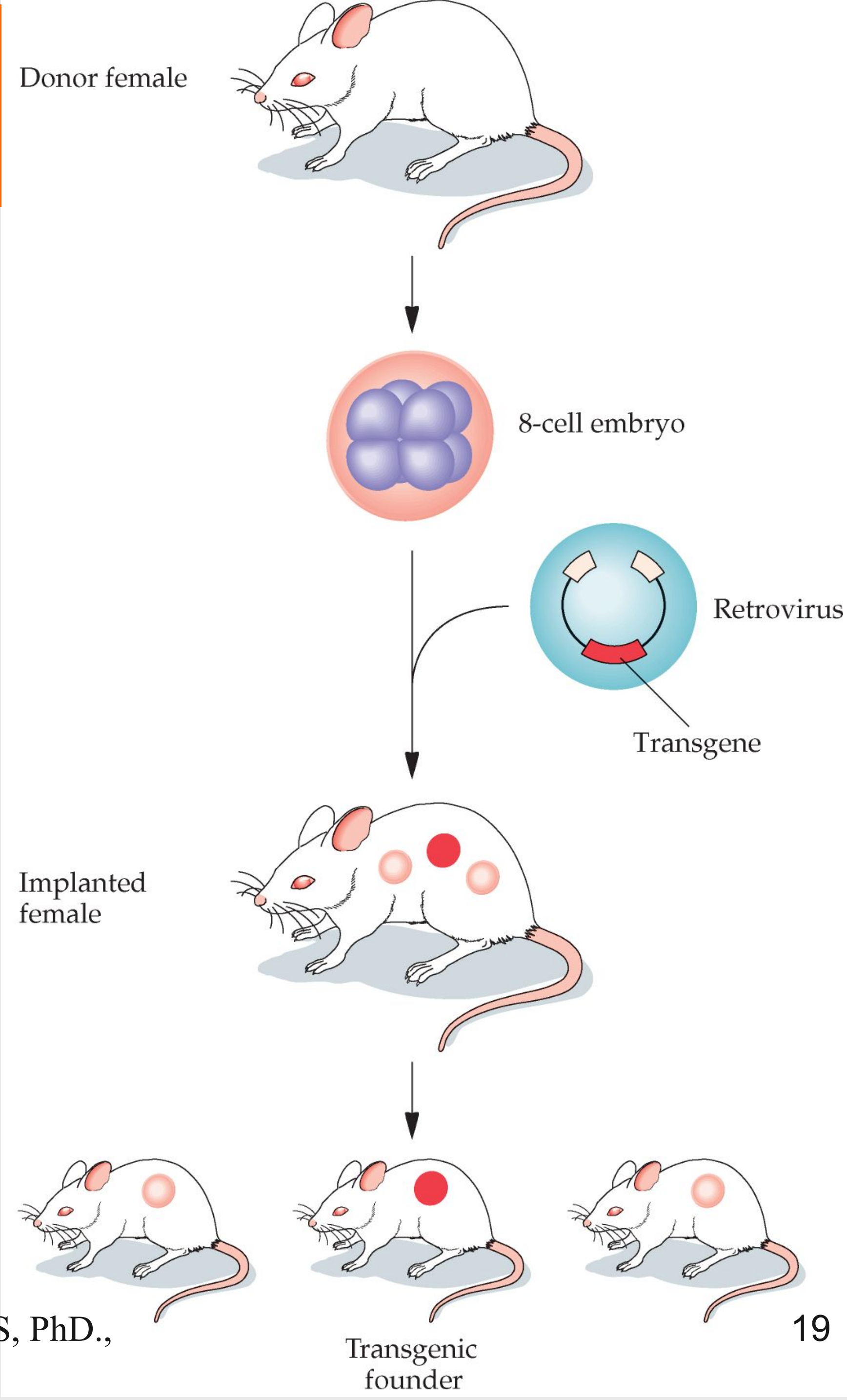
DNA microinjection



Retrovirus-mediated gene transfer

Retroviral vectors can be used to create transgenic animals

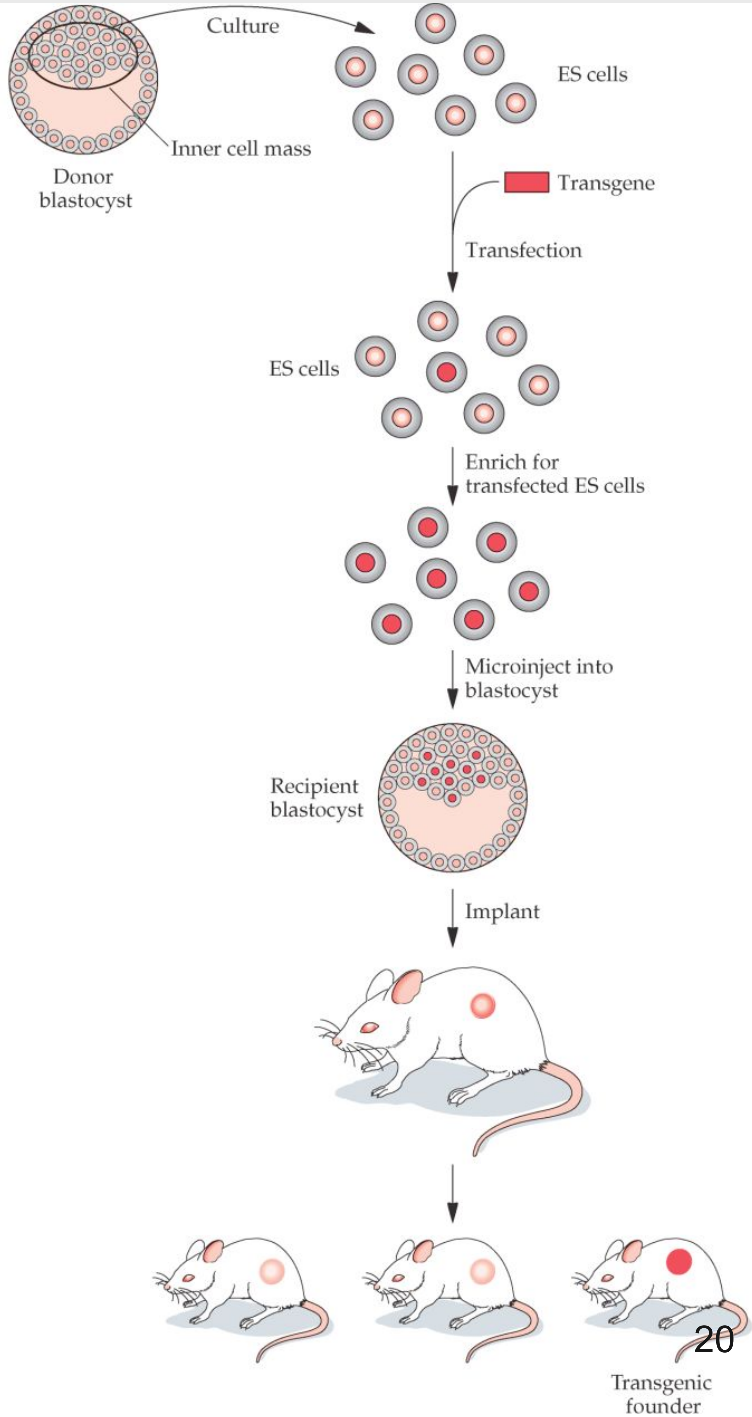
- Infecting mouse embryo with a retrovirus which carry the new gene.
- Using virus as a vector .



Embryonic stem cell-mediated gene transfer

Genetically engineered embryonic stem (ES) cells can be used to create transgenic animals, but this method is labor intensive and used to allow for gene targeting via homologous recombination.

- The blastocyst (inner layer of a fertilized egg) is harvested and mixed with recombinant DNA and inserted back in the blastocyst.
- Involves:
  - Isolation of totipotent stem cells (stem cells that can develop into any type of specialized cell) from embryos.
  - The desired gene is inserted into these cells.
  - Cells containing the desired DNA are incorporated into the host's embryo.



Embryonic stem cell-mediated gene transfer

Embryonic stem cells used to create mice with altered genome

Changes in the germ line -> transgenic offspring!!!

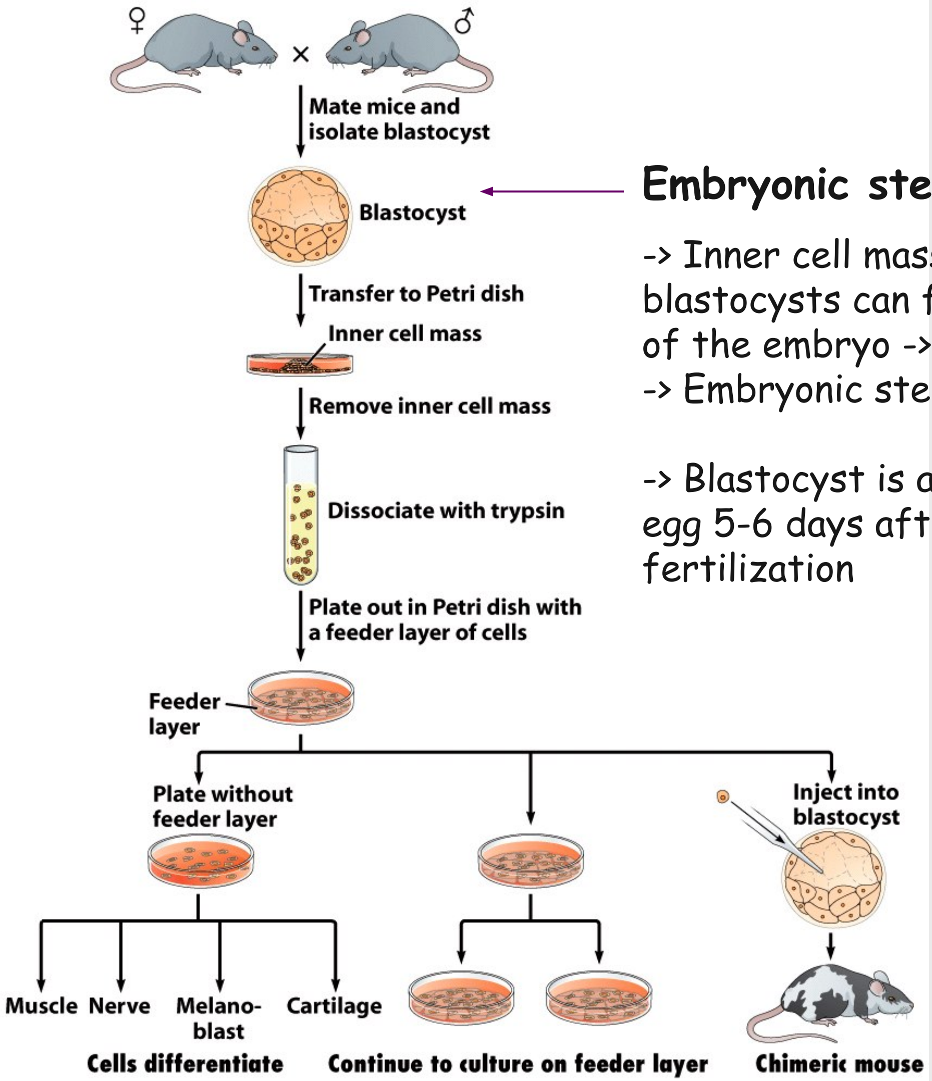


Figure 6-16
Genes and Genomics: A Short Course (3e)

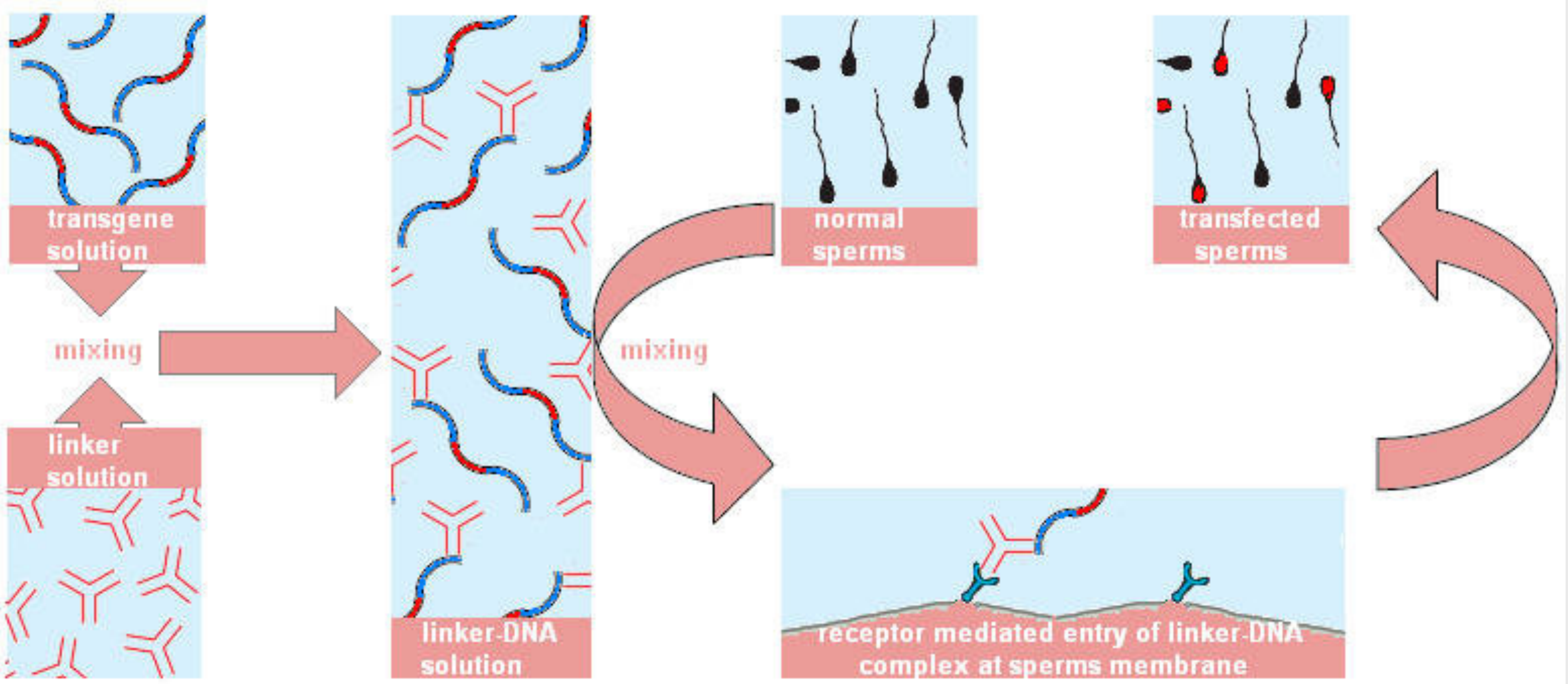
Embryonic stem cells

-> Inner cell mass (ICM) of blastocysts can form all cells of the embryo -> Pluripotent -> Embryonic stem cells

-> Blastocyst is a fertilized egg 5-6 days after fertilization

Sperm-mediated transfer

- Use of “Linker protein" to attach DNA to sperm which transfer the new DNA during fertilization.



Gene-gun (Shutgun)

Used for animal and plant cells

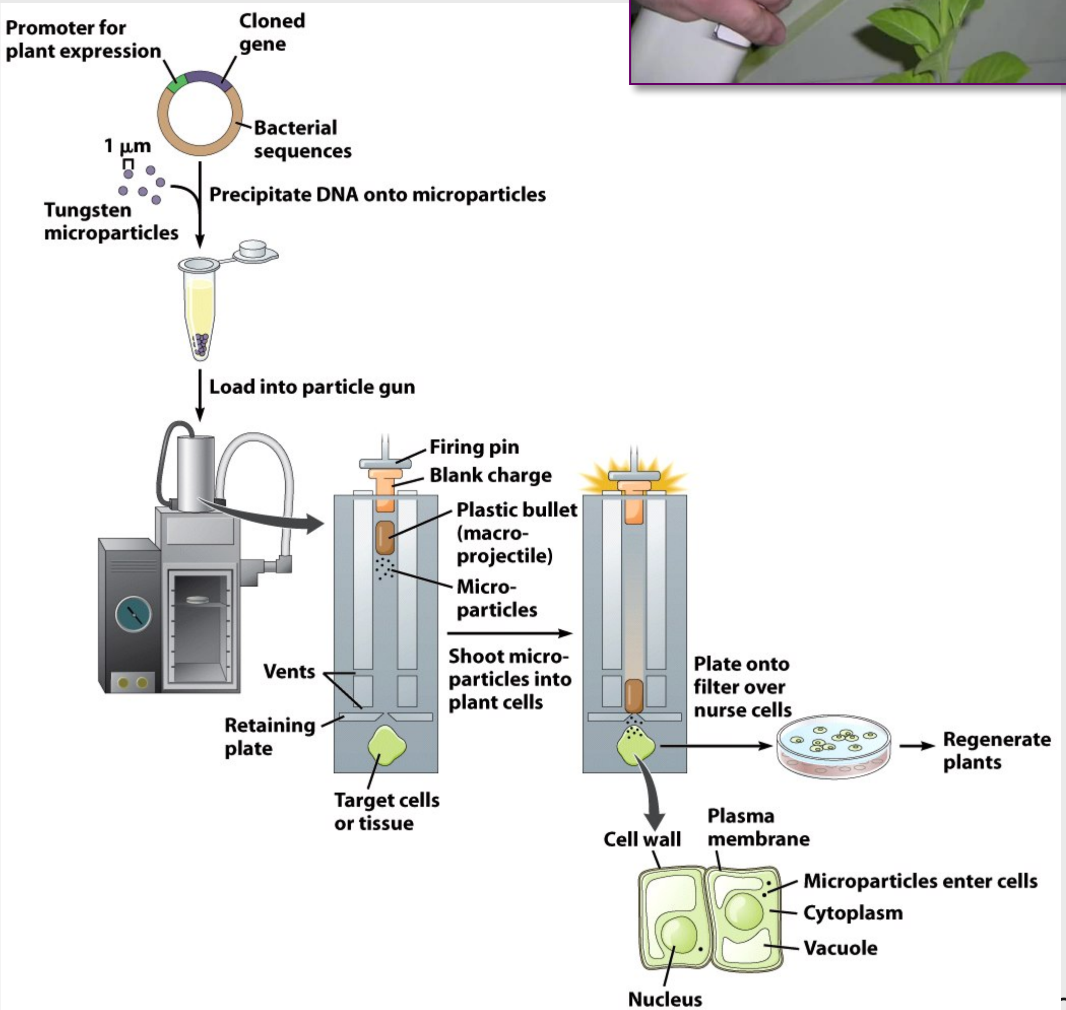
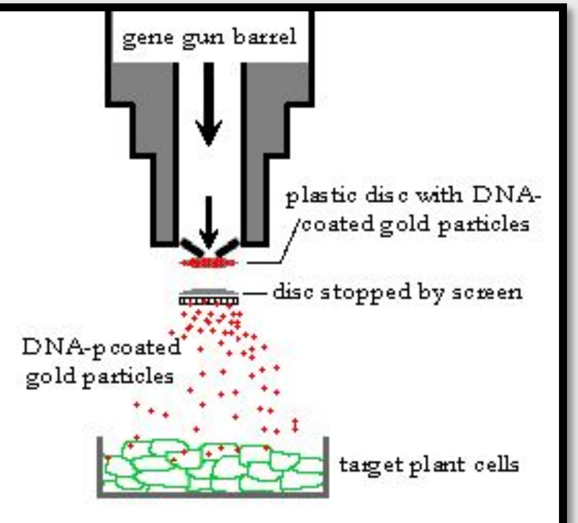


Figure 6-11
Genes and Genomics: A Short Course (3e)



Gene-gun (Shutgun)

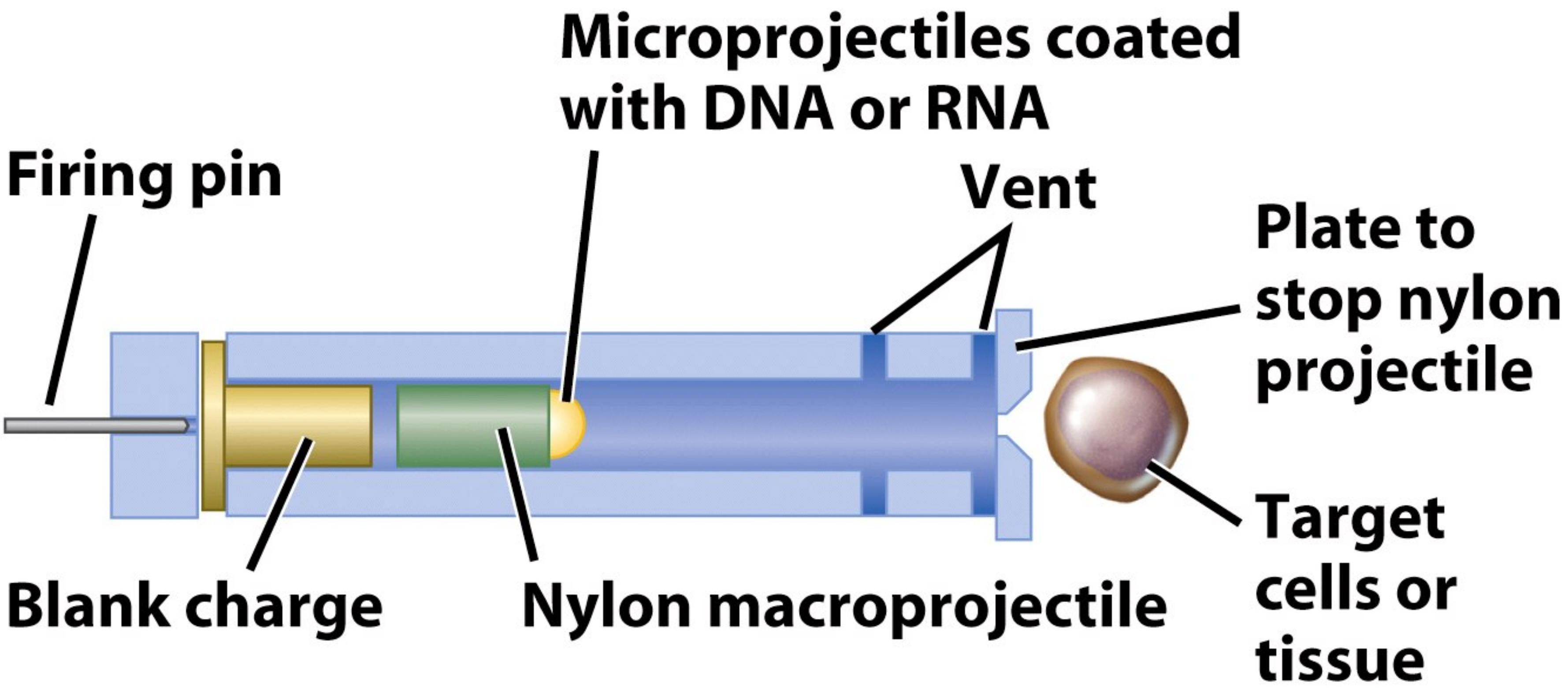


Figure 31-3 Brock Biology of Microorganisms 11/e

Electroporation

Used for:
-> Bacteria
-> Fungi
-> mammalian cells
-> some plant cells

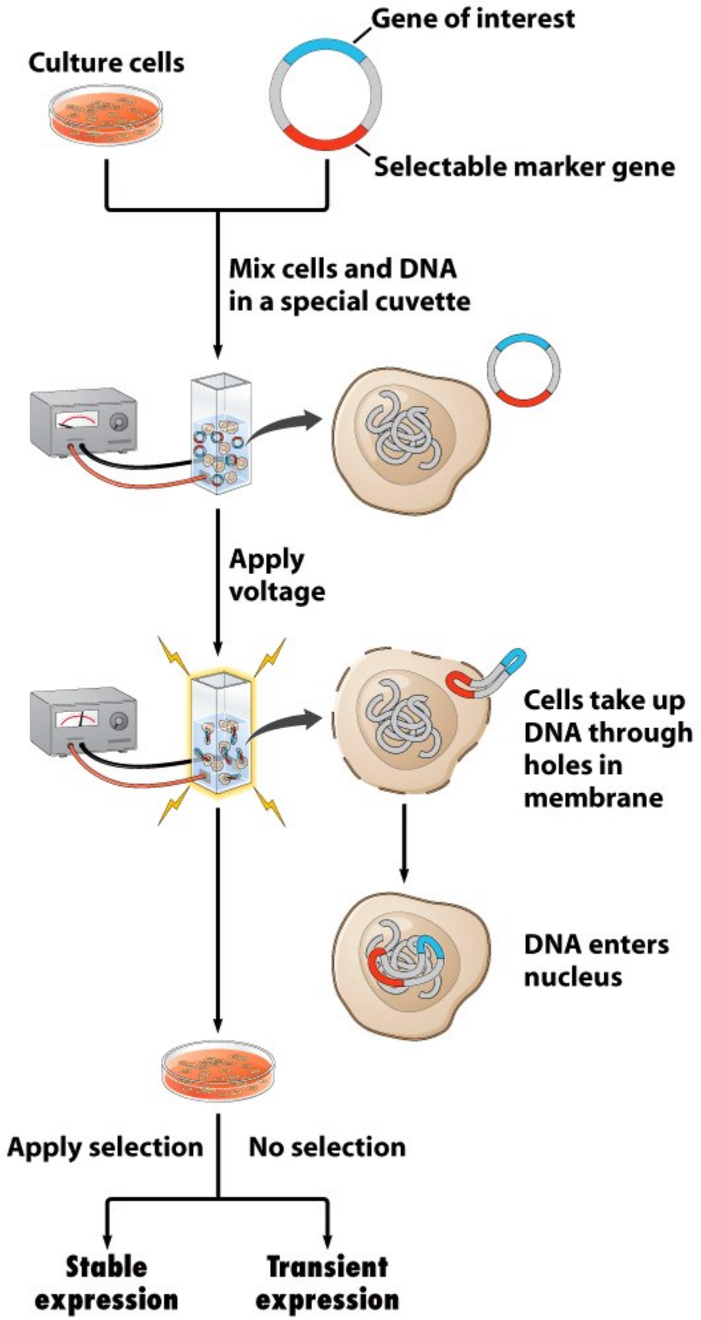
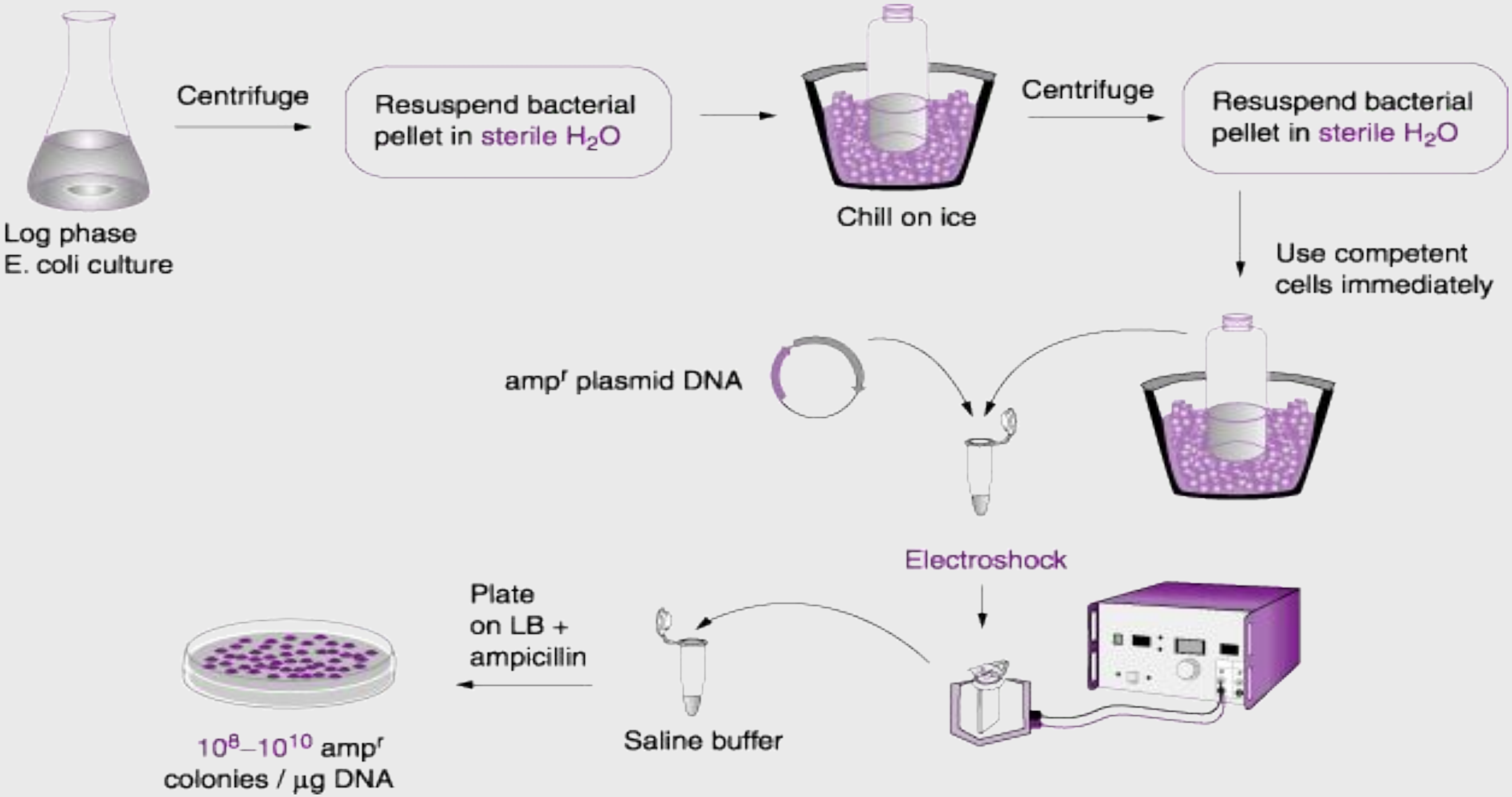


Figure 6-8
Genes and Genomics: A Short Course (3e)

Electroporation



Lipofection

Used mainly for animal cells

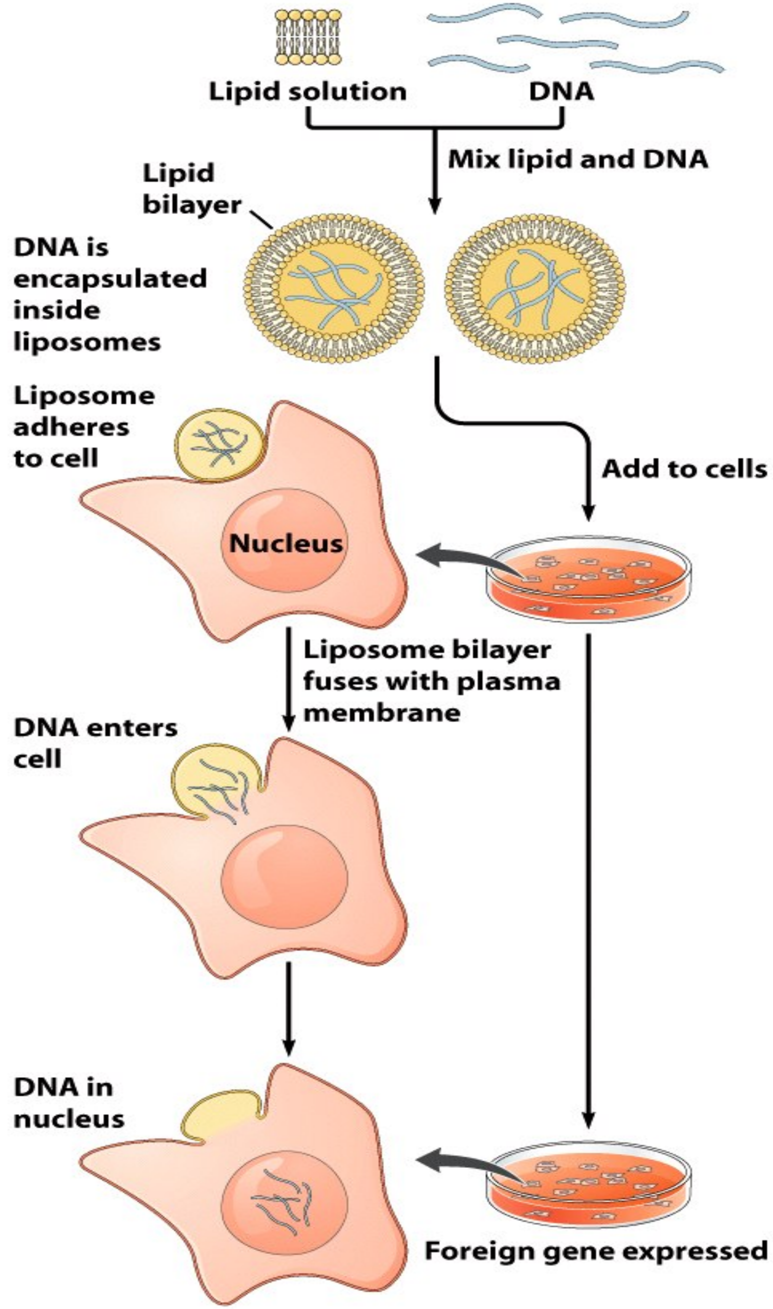


Figure 6-9
Genes and Genomics: A Short Course (3e)

Transgenics and Its Applications

- **“Pharm” animals (transgenic livestock)**
  - Bioreactors whose cells have been engineered to synthesize marketable proteins
  - DNA constructs contain desired gene and appropriate regulatory sequences (tissue-specific promoters)
  - More economical than producing desired proteins in cell culture

APPLICATIONS OF TRANSGENICS

- The benefits of Transgenic animals to human welfare can be grouped into areas:
  - Agriculture
  - Medicine
  - Industry

Medical Applications of Transgenics

A) Xenotransplantation

- Transplant organs may soon come from transgenic animals.

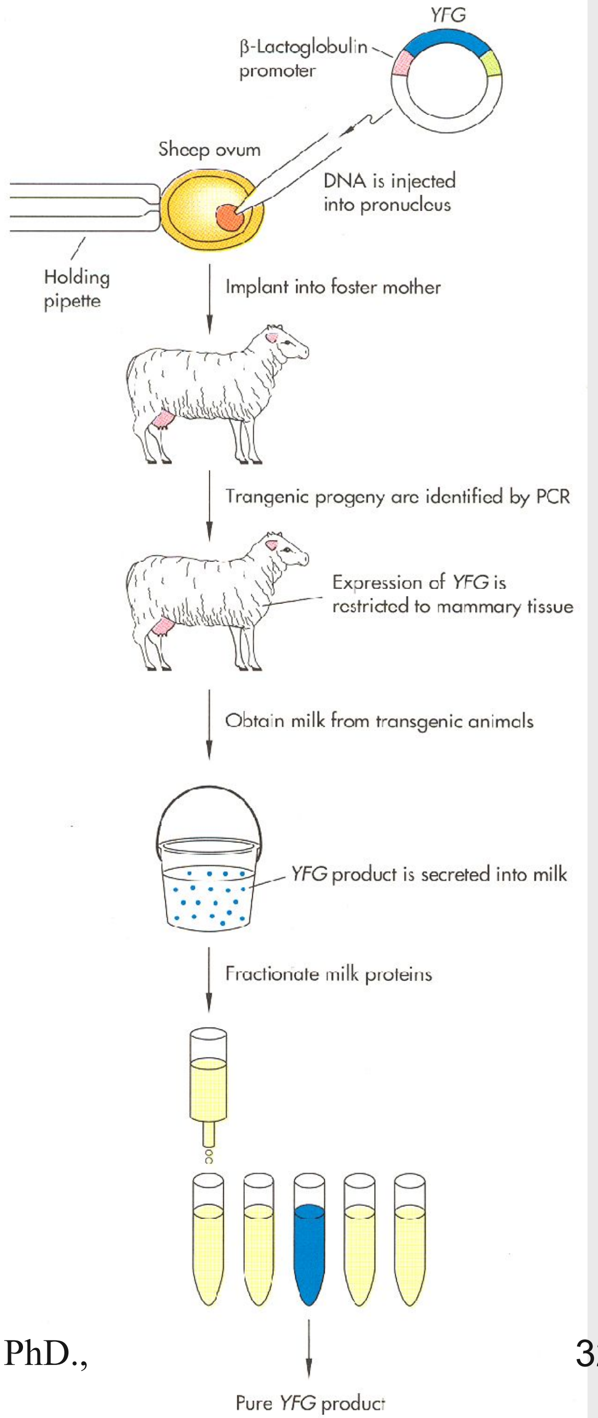
Medical Applications

B) Nutritional supplements and pharmaceuticals

- Products such as insulin, growth hormone, and blood anti-clotting factors may soon be or have already been obtained from the milk of transgenic cows, sheep, or goats.
- The first transgenic cow (Rosie) produced human protein-enriched milk at 2.4 grams per liter.
- This transgenic milk is a more nutritionally balanced product than natural milk and could be given to babies or the elderly with special nutritional or digestive needs.

Transgenic cattle, sheep, goats, and pigs

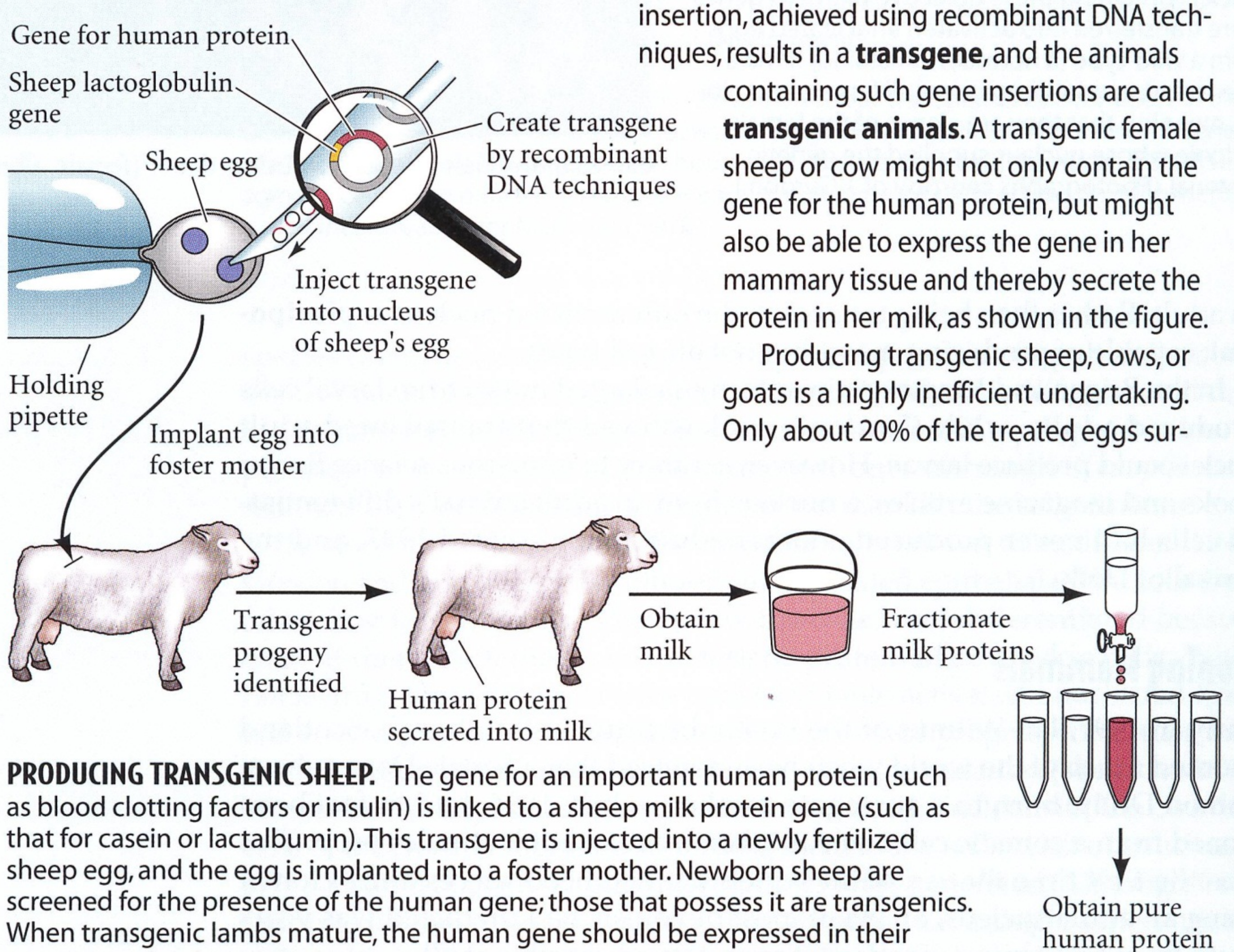
- Using the mammary gland as a bioreactor (see adjacent figure)
- Increase casein content in milk
- Express lactase in milk (to remove lactose)
- Resistance to bacterial, viral, and parasitic diseases



Transgenic Cattle, Sheep, Goat, Pigs

insertion, achieved using recombinant DNA techniques, results in a **transgene**, and the animals containing such gene insertions are called **transgenic animals.** A transgenic female sheep or cow might not only contain the gene for the human protein, but might also be able to express the gene in her mammary tissue and thereby secrete the protein in her milk, as shown in the figure.

Producing transgenic sheep, cows, or goats is a highly inefficient undertaking. Only about 20% of the treated eggs sur-



PRODUCING TRANSGENIC SHEEP. The gene for an important human protein (such as blood clotting factors or insulin) is linked to a sheep milk protein gene (such as that for casein or lactalbumin). This transgene is injected into a newly fertilized sheep egg, and the egg is implanted into a foster mother. Newborn sheep are screened for the presence of the human gene; those that possess it are transgenics. When transgenic lambs mature, the human gene should be expressed in their mammary glands and the human protein will be secreted in their milk. From the milk one can then isolate large amounts of the protein for pharmaceutical use.

Production of pharmaceutical proteins -> drugs

Problems:

Highly inefficient

Only 20% of the eggs survive and only 5% of them produce product

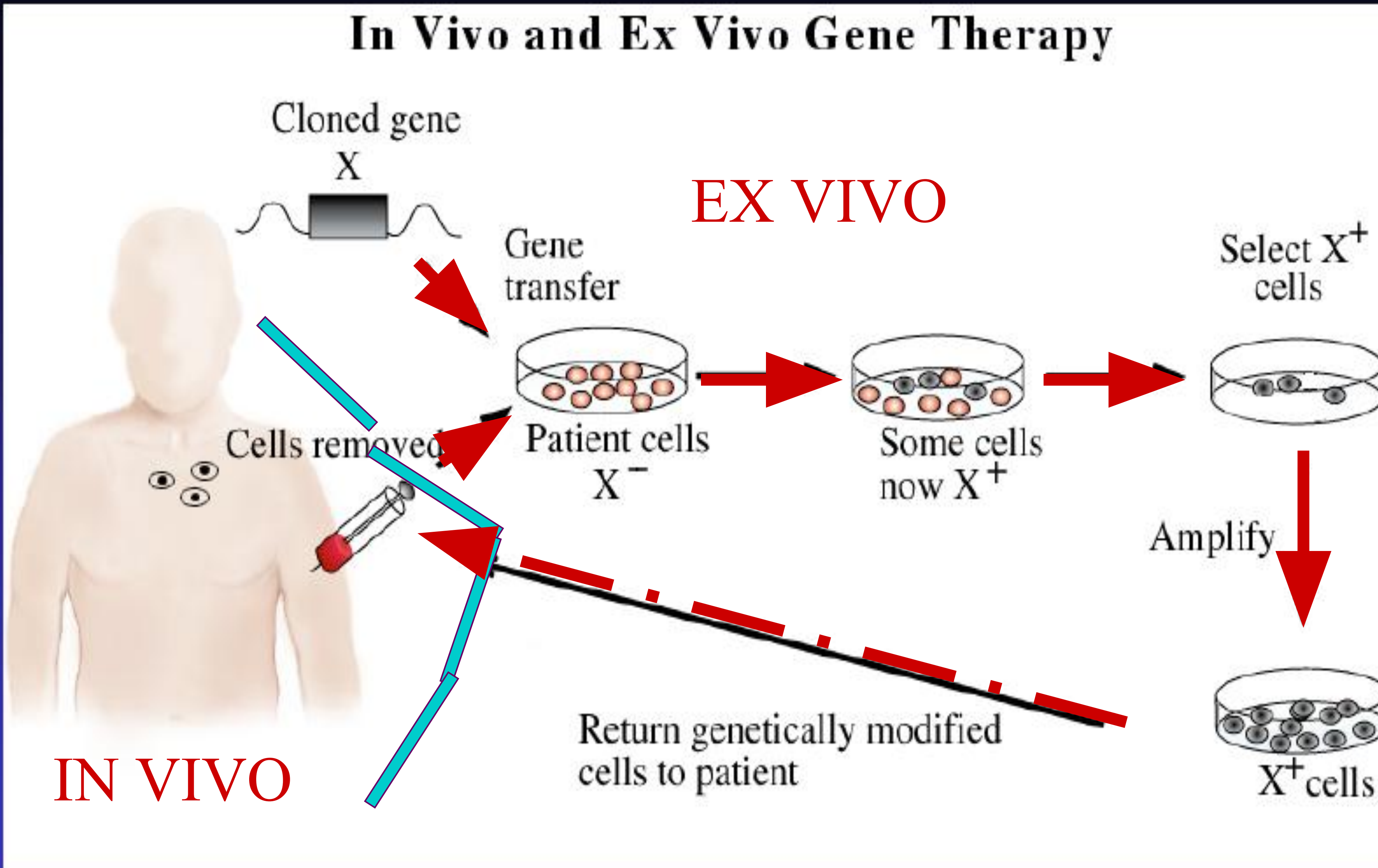
Some human proteins expressed in the mammary glands of transgenic animals:

- Erythropoietin
- Factor IX
- Factor VIII
- Fibrinogen
- Growth hormone
- Hemoglobin
- Insulin
- Monoclonal antibodies
- Tissue plasminogen activator (TPA)
- α1-antitrypsin
- Antithrombin III (the first transgenic animal drug, an anticlotting protein, approved by the FDA in 2009)

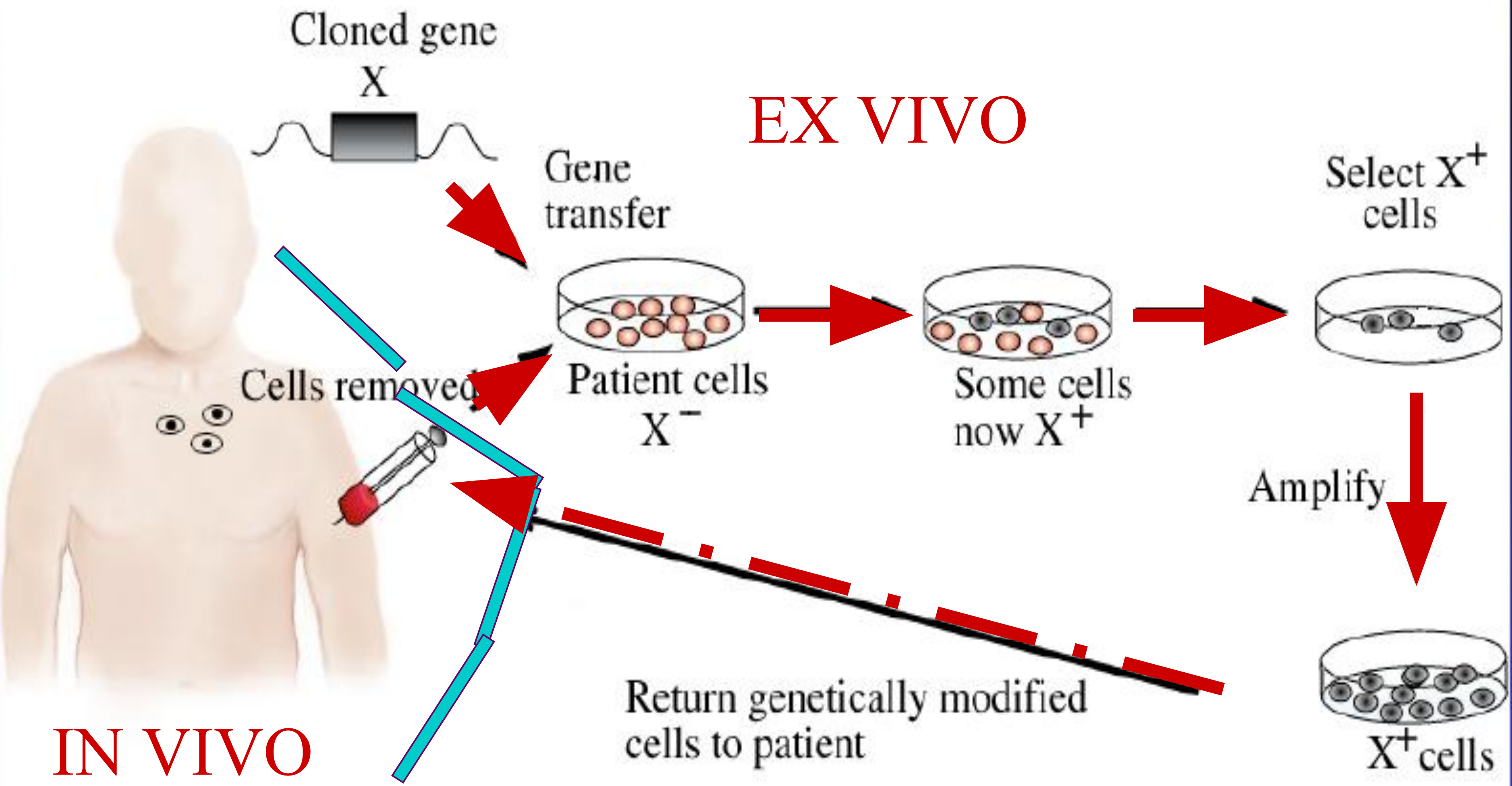
Medical Applications

C) Human gene therapy

- Human gene therapy involves adding a normal copy of a gene (transgene) to the genome of a person carrying defective copies of the gene.


In Vivo and Ex Vivo Gene Therapy
Cloned gene
X
EX VIVO
Gene transfer
Select X^+ cells
Cells removed
Patient cells X^-
Some cells now X^+
Amplify
Return genetically modified cells to patient
IN VIVO
X^+ cells

In Vivo and Ex Vivo Gene Therapy



Genomic and cDNA Libraries

Genomic and cDNA Libraries:

- A DNA library is a collection of cloned restriction fragments of the DNA of an organism.
- Two kinds of libraries are commonly used:
  - Genomic libraries and
    - Contain a copy of every DNA nucleotide sequence in the genome
  - Complementary DNA (cDNA) libraries.
    - Contain those DNA sequences that only appear as processed mRNA molecules and these differ from one cell type to another.
- Note: cDNA lacks introns and the control regions of the genes, whereas these are present in genomic DNA.

Genomic libraries

- Genomic library:
  - A cloned set of rDNA fragments representing the entire genome of an organism
- Method of creation:
  - The total DNA of the organism digested with a restriction endonuclease and subsequent ligation to an appropriate vector
  - The recombinant DNA molecules replicate within host bacteria.
  - The amplified DNA fragments thus represent the entire genome of the organism and are called a genomic library.

cDNA libraries

- cDNA libraries:
  - Small subsets of sequences found in a genomic library
  - mRNA represents genes that are actively transcribed (or expressed) at any given time in a particular cell type
- Method of creation:
  - mRNA Converted into a DNA copy (=cDNA) using a series of enzymatic reactions and oligonucleotides
     Primer, reverse transcriptase, DNA polymerase I, S1 nuclease, linkers, restriction enzymes, vector are used.
  - Size of library depends on abundance of message

Read more....


Thank you...